

Example 1.3. The map $|\cdot|: k \rightarrow \mathbb{R}_{\geq 0}$ defined by

$$|x| = \begin{cases} 1 & \text{if } x \neq 0, \\ 0 & \text{if } x = 0, \end{cases}$$

is the *trivial absolute value* on k . It is nonarchimedean.

Lemma 1.4. An absolute value $|\cdot|$ on a field k is nonarchimedean if and only if

$$|\underbrace{1 + \cdots + 1}_n| \leq 1$$

for all $n \geq 1$.

Proof. See Problem Set 1. □

Corollary 1.5. In a field of positive characteristic every absolute value is nonarchimedean, and the only absolute value on a finite field is the trivial one.

Proof. Suppose k has positive characteristic p . Then every $n = 1 + \cdots + 1$ lies in the prime field $\mathbb{F}_p$ of k and satisfies $n^p = n$. This implies $|n|^p = |n|$, so $|n|$ is either 0 or 1 and k is nonarchimedean (by the lemma). If k is finite, say of cardinality q , then $x^q = x$ for all $x \in k$ and $|x|^q = |x|$ is 0 or 1, with the former occurring only for $x = 0$. □

Definition 1.6. Two absolute values $|\cdot|$ and $|\cdot|'$ on the same field k are *equivalent* if there exists an $\alpha \in \mathbb{R}_{>0}$ for which $|x|' = |x|^\alpha$ for all $x \in k$.

1.3 Absolute values on $\mathbb{Q}$

To avoid confusion we will denote the usual absolute value on $\mathbb{Q}$ (inherited from $\mathbb{R}$) by $|\cdot|_\infty$; it is an archimedean absolute value. But there are infinitely many others. Recall that any element of $\mathbb{Q}^\times$ may be written as $\pm \prod_q q^{e_q}$, where the product ranges over primes and the exponents $e_q \in \mathbb{Z}$ are uniquely determined (as is the sign).

Definition 1.7. For a prime p the *p-adic valuation* $v_p: \mathbb{Q} \rightarrow \mathbb{Z}$ is defined by

$$v_p\left(\pm \prod_q q^{e_q}\right) := e_p,$$

and we define $v_p(0) := \infty$. The *p-adic absolute value* on $\mathbb{Q}$ is defined by

$$|x|_p := p^{-v_p(x)},$$

where $|0|_p = p^{-\infty}$ is understood to be 0.

Theorem 1.8 (OSTROWSKI'S THEOREM). Every nontrivial absolute value on $\mathbb{Q}$ is equivalent to $|\cdot|_p$ for some $p \leq \infty$.

Proof. See Problem Set 1. □

Theorem 1.9 (PRODUCT FORMULA). For every $x \in \mathbb{Q}^\times$ we have

$$\prod_{p \leq \infty} |x|_p = 1.$$

Proof. See Problem Set 1. □